Abstract

While recent integrations of large language models (LLMs) with ROS 2 enable natural language robot interaction, command-processing latencies and narrow-space navigation constraints—exemplified by recent benchmarks and the BARN 2026 challenge—underscore the necessity of robust, verifiable physical execution. Direct language-to-goal dispatch remains fundamentally unreliable due to linguistic ambiguity, spatial hallucinations of unregistered coordinates, stale sensor states, and ungrounded task completion claims. To bridge this gap, this study proposes an evidence-grounded indoor navigation and task decision AI agent deployed on a differential-drive TurtleBot3 Burger platform. The system decouples high-level semantic reasoning from deterministic real-time control by integrating schema-constrained LLM intent parsing, a registered landmark registry, pre-dispatch state and transform (TF) validation, whitelisted Nav2 tool calling, and persistent event audit logging. The planned evaluation protocol encompasses 60 Chinese natural language commands categorized into 20 development and 40 held-out test cases across six semantic templates, four simulated obstacle navigation scenarios (40 runs per method), and 12 paired physical robot test cases, comparing the full agent against fixed-menu (B0) and unconstrained LLM (B1) baselines alongside ablation variants. In strict adherence to proposal reporting standards without fabricated empirical data, the proposed framework is anticipated to mitigate improper dispatches from ambiguous inputs, guarantee execution traceability through verifiable status reports, and characterize the latency trade-offs between interactive reasoning and mobile robot navigation.

Index Terms—Autonomous mobile robots, TurtleBot3 Burger, Large Language Models (LLMs), Nav2, indoor navigation, task decision agent, tool calling, failure traceability.